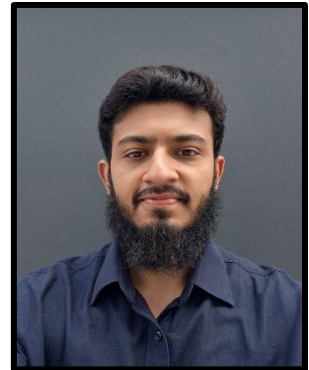


AL FARABI

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EDUCATION

Bachelor of Science (B.Sc.) in Mechanical Engineering

Jan 2022 - Present

Bangladesh University of Engineering and Technology (BUET), Dhaka
CGPA: 3.63/4.00 (after 7 semesters)

Higher Secondary Certificate (HSC)

June 2020

Jhenaidah Cadet College
GPA: 5.00

Secondary School Certificate (SSC)

March 2018

Jhenaidah Cadet College
GPA: 5.00

RESEARCH EXPERIENCE

Undergraduate Researcher

April 2025 – Present

Aman Uddin Lab (AUL)

Supervisor: Dr. Md. Aman Uddin, Associate Professor, Dept. of Mechanical Engineering, BUET

**Thesis: Green Hydrogen and EV Integration in Bangladesh's Low-Carbon Energy Transition:
A Sector-Coupled Multi-Tool Modeling Approach.**

Energy system is optimized by PyPSA; EV integration and spatial distribution of power generation and transmission is achieved.

- **Energy System Modelling**
- **Motor Efficiency and Performance Analysis**
- **Thermal Management and Heat Exchanger Design**
- **Structural Analysis and Simulation**

PROJECTS

1. Efficiency Eye: A Data Acquisition System for Measuring the Efficiency of Motors

Designed an Arduino based DAQ framework to evaluate motor efficiency by acquiring real-time performance data, aiming to optimize industrial power consumption.

2. Green-Cooling: Efficient Thermal Management for Tomorrow's Data Needs

Fabricated a double-pipe heat exchanger and cooling tower prototype, performing thermodynamic calculations to optimize data center thermal management.

3. Rocket Booster Catcher Mechanism Analysis

Modeled a rocket booster catcher mechanism in SolidWorks and simulated stress distribution using Ansys.

4. Comparative Study of Genetic Algorithm and Ziegler–Nichols PID Tuning for Aircraft Longitudinal Dynamics

Conducted a comparative analysis of Genetic Algorithm (GA) and Ziegler–Nichols PID tuning methods to optimize longitudinal flight control, evaluating stability and performance response for aircraft dynamic system.

TECHNICAL SKILLS

- **Programming and Data Analysis**
Languages: C, Python, MATLAB, Arduino
- **Technical and Simulation Software**
SolidWorks, Autocad, Ansys, PyPSA
- **Presentation, Documentation and Writing**
Latex, MS Office, MS Powerpoint, Canva

PROFESSIONAL EXPERIENCE

Intern, ACI Pharmaceutical Limited.

April 2025 – May 2025

- Analyzed the operational mechanics of pharmaceutical facility infrastructure, gaining practical insights into industrial HVAC controls and climate management.
- Monitored power generation technologies and energy management systems to understand power optimization and efficiency in large-scale manufacturing.

Awards & Honors

- University Merit Scholarship
- Board Scholarships: HSC (2020), SSC (2018), JSC (2016)

REFERENCES

1. Dr. Muhammed Mahbubur Razzaque

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Bangladesh University of Engineering and Technology
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2. Dr. Aman Uddin

B.Sc. Supervisor

Associate Professor, Department of Mechanical Engineering
Bangladesh University of Engineering and Technology
Email: amanuddin@me.buet.ac.bd